# DEVOPS with MULTI-CLOUD
# Practice Tasks

**Institute Name :** V Cube software solutions
**Course :** DevOps with Multi-Cloud
**Batch :** 30
**Trainer :** Krishna reddy sir

**Prepared by :** G.Bhavish
(MCD-AZ30-024)

# TASK-21 : Virtual Private Network (VPN).

**Date** : 18/02/26

**Objective :-**

To securely connect on-premises networks, remote users, or other virtual networks to Azure Virtual Network using encrypted communication over the internet, ensuring safe and reliable data transfer.

**VPN :-**

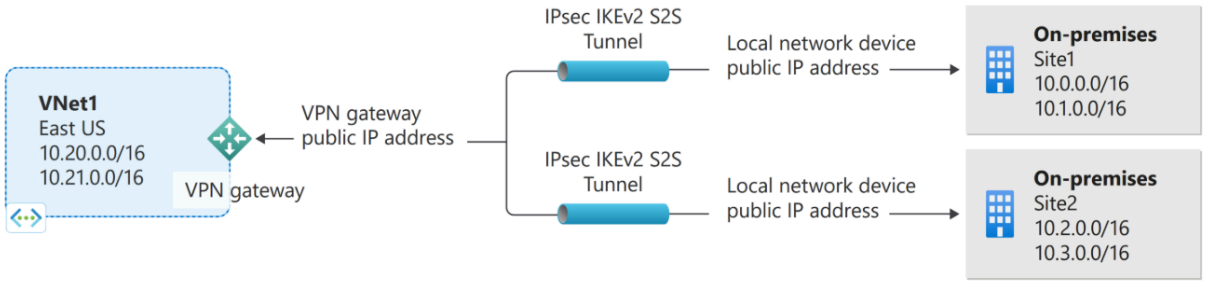
→ To create a VPN we need to have a dedicated subnet.
→ it creates a secure tunnel between the network.
→ Azure VPN is a secure network connection that connects an on premises network or users to a virtual network using encrypted tunnels over the internet.
→ Azure vpn uses vpn gateway to create secure connections.
→ It encrypts traffic using ipsec/ike protocol
→ it works at OSI Layer 3 (Network layer).

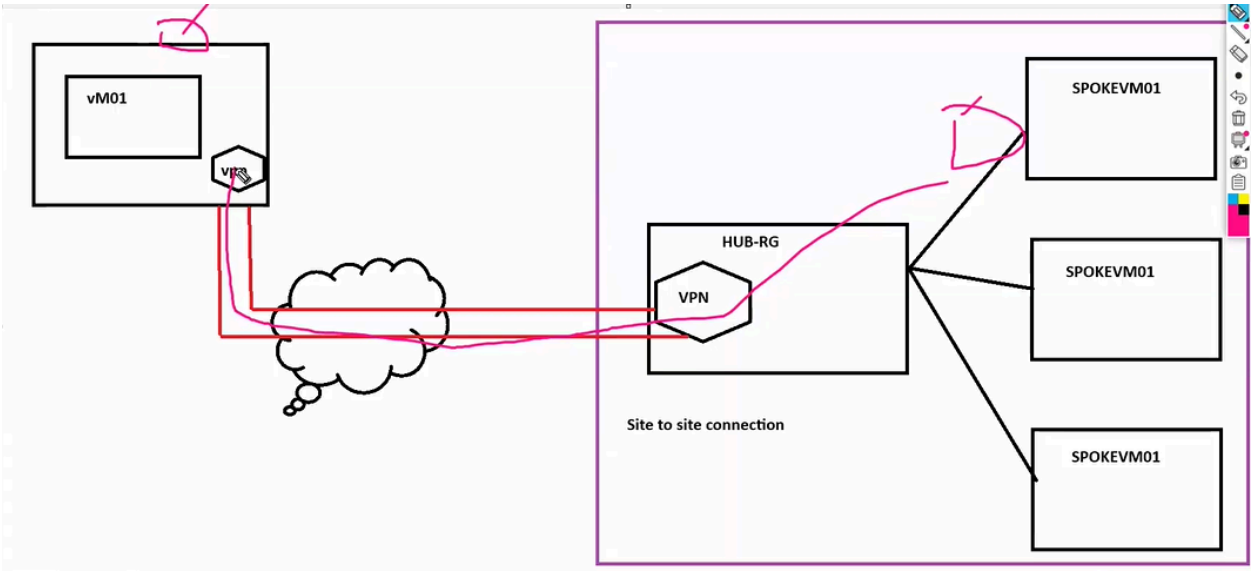
**Types of Vpn Connections :-**

- Site to Site
- Point to Site
- Vn to Vn

➤ <u>Site to Site</u> :-

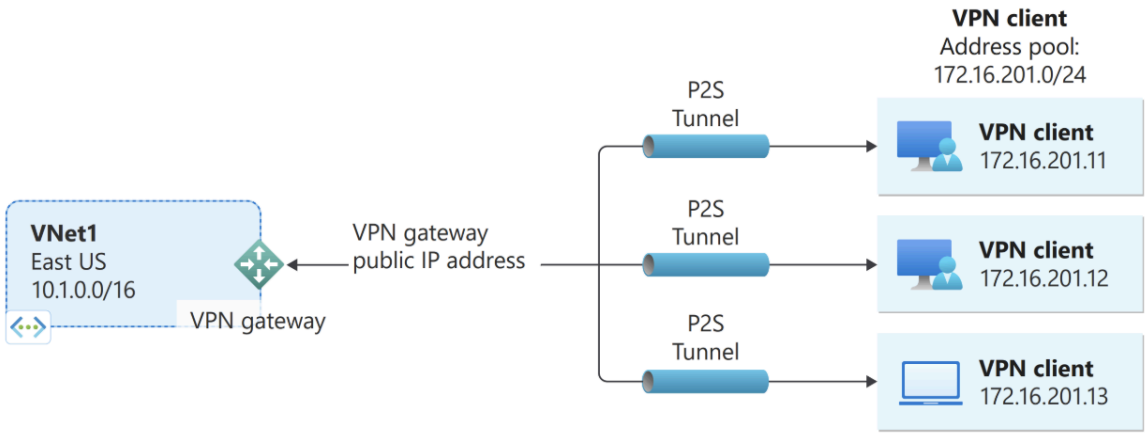
- The site to site vpn connects one entire network to another network over the internet.
- When s2s is configured, by default a tunnel will be created from one branch to another.
- The data which will be passing through the tunnel will be encrypted and no one can access the data.

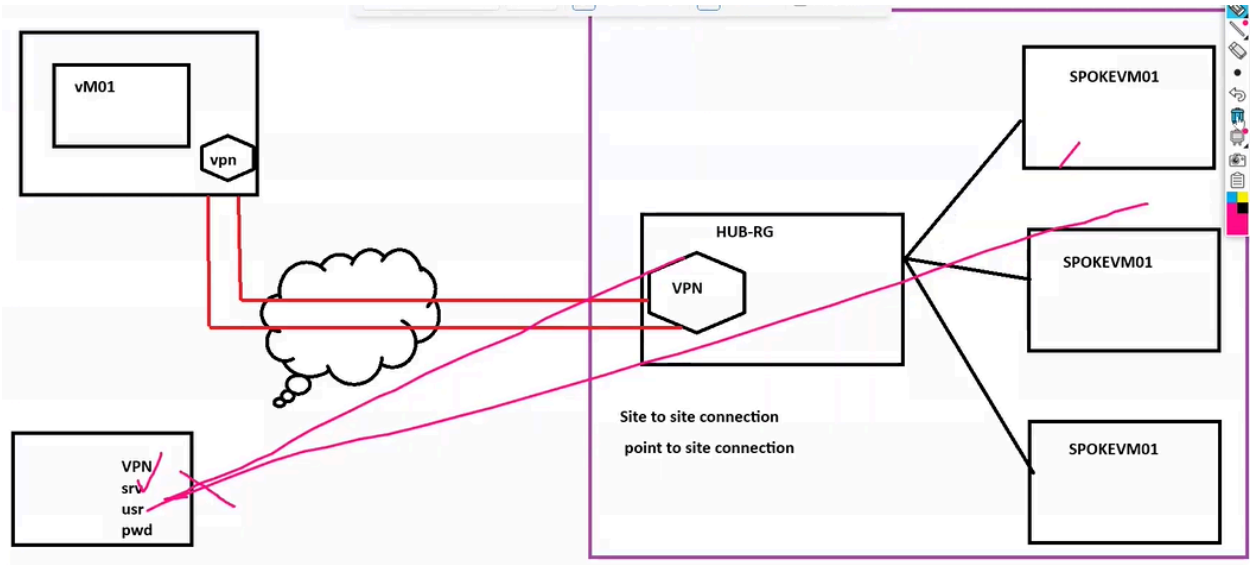


➤ <u>Point to Site</u> :-

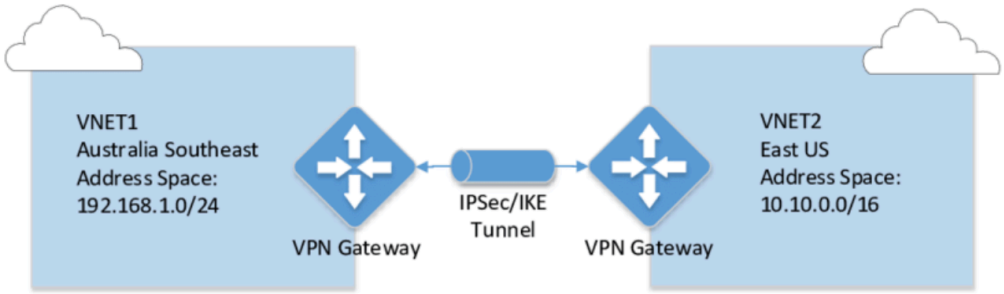
- The point to site vpn connects one individual computer (user) to an Azure network.
- It is on demand connection
- We can connect whenever we want.
- Used for remote employees or admins.



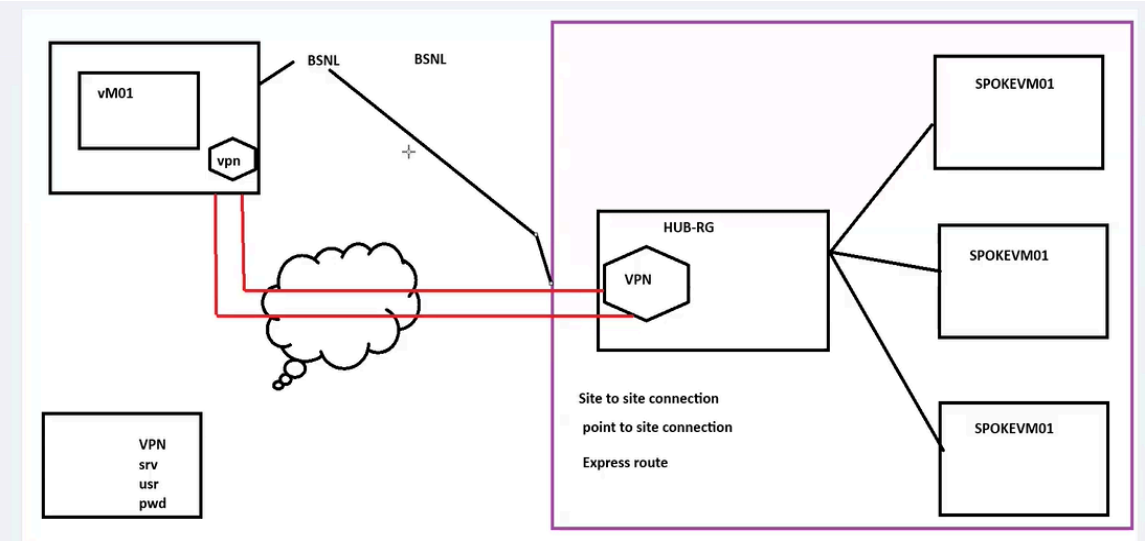
➢ <u>Vn to Vn</u> :-

- Used when VNets are in different regions.
- Uses IPsec encrypted tunnel.
- Each VNet must have its own VPN Gateway.
- Works even if VNets are in different subscriptions.




→ <u>Express route</u> :-

- We will have a physical cable connection.
- The cable connection is established based on the partners network like bsnl, jio, airtel. These cables are already available underground.

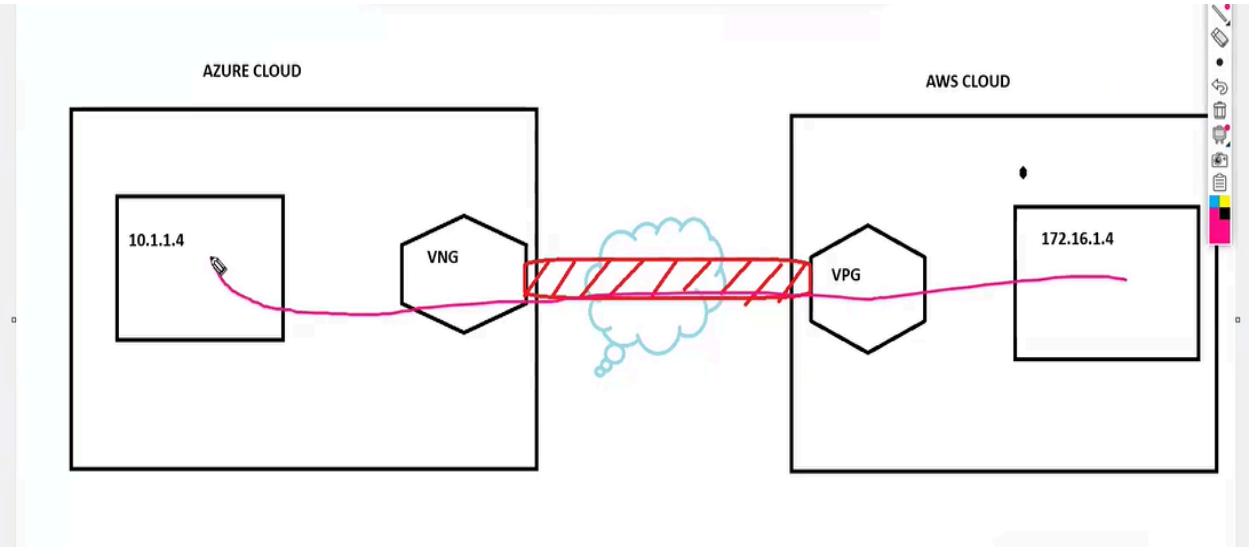


- The connection goes like(from hyd-chennai):
    - our office _to_ nearest bsnl office(hyd).
    - hyd bsnl office _to_ chennai bsnl office.
    - chennai bsnl office _to_ data center(chennai).

- In the express route we can transfer huge amounts of data, most of the companies use the express route.

⦿ <u>Site to site</u> :-

For the site to site connection we are going to use two clouds (aws & azure).




→ we are going to create two vm's, one in aws and one in azure

→ these two machines will communicate privately through the tunnel over the internet.

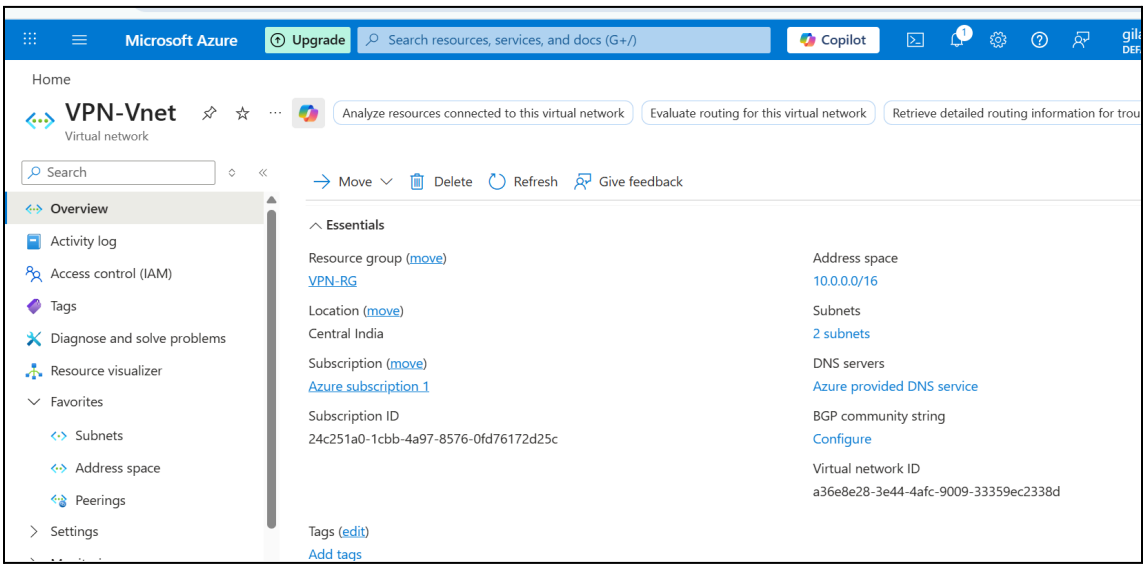
In Azure :-

→ Create resource group & virtual network

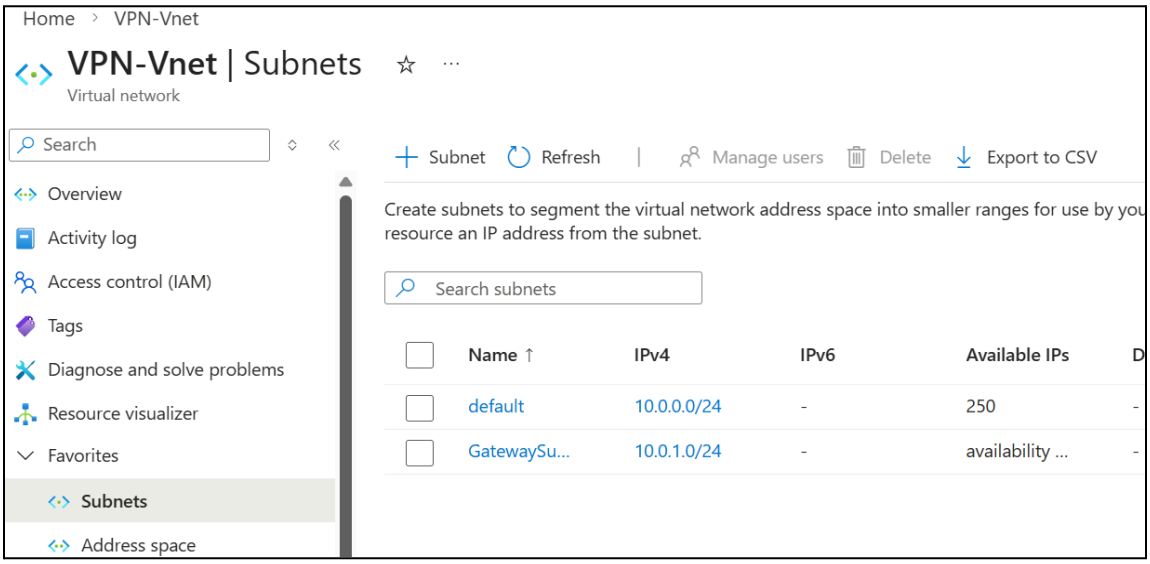
Vpn-rg , vpn-vn

→ Create a dedicated subnet

Vpn-vn > subnets > +sn > purpose > VNG.




fig(1) created vpn virtual network.




fig(2) created a dedicated subnet for VNG.

→ Create a virtual network gateway (VNG).

Search vng > vpn gateway > +create vpn gateway

Name – vpn01

Gateway configuration :-

Gateway type – vpn

Sku – VPNGW1AZ

Generation – geneation1

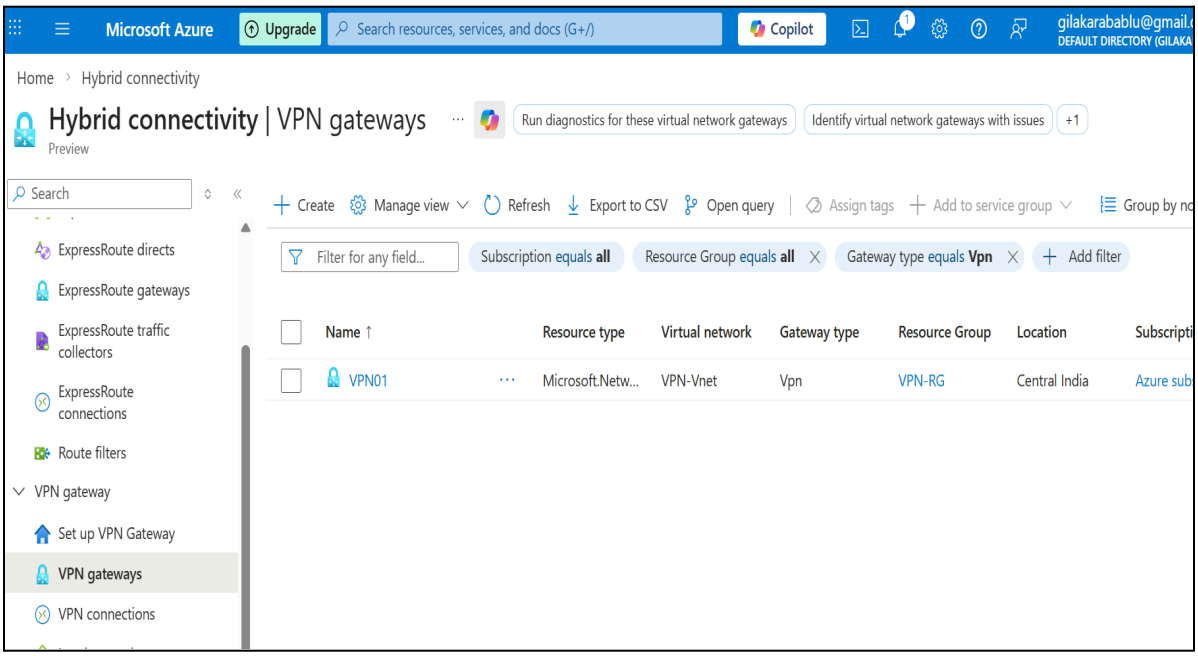
Vn – vpn-vn

Public ip address :-

Pub-ip — create new

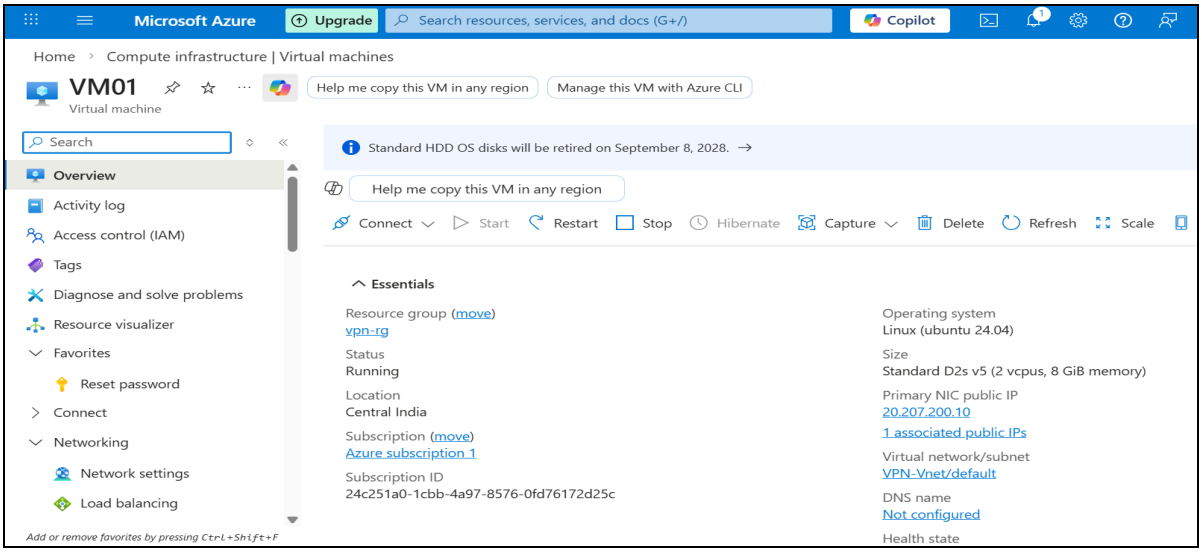
Pub ip address name – vpn-pubip

Review+create ✔️




fig(3)created a vng i.e vpn-gateway.

→ Now create a vm.




fig(4) created a vm.

## In Aws :-

→ First create a vpc.

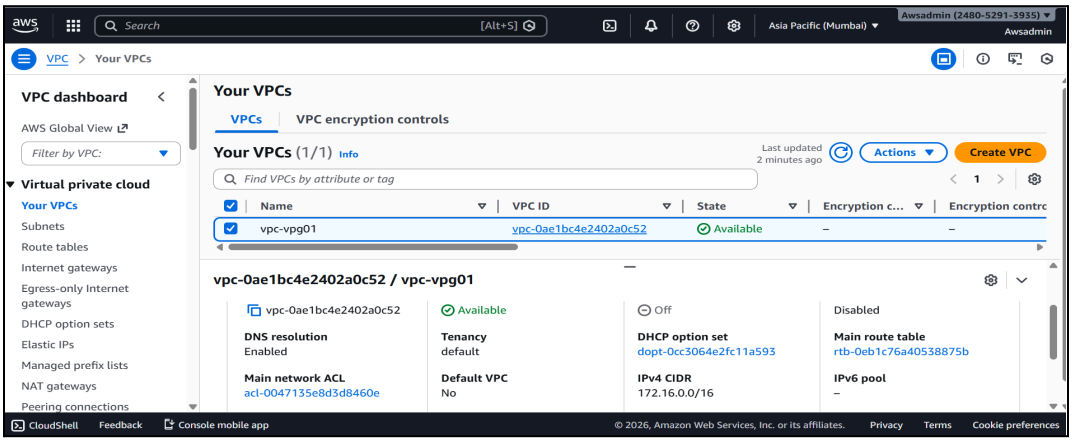
Search vpc > vpc > create vpc

Resource to create – vpc only

Name – vpg-vpc01

Ipv4 – 172.16.0.0/16

Create vpc ✔️




fig(5) created a vpc

→ vpc is created now, copy the vpc id .

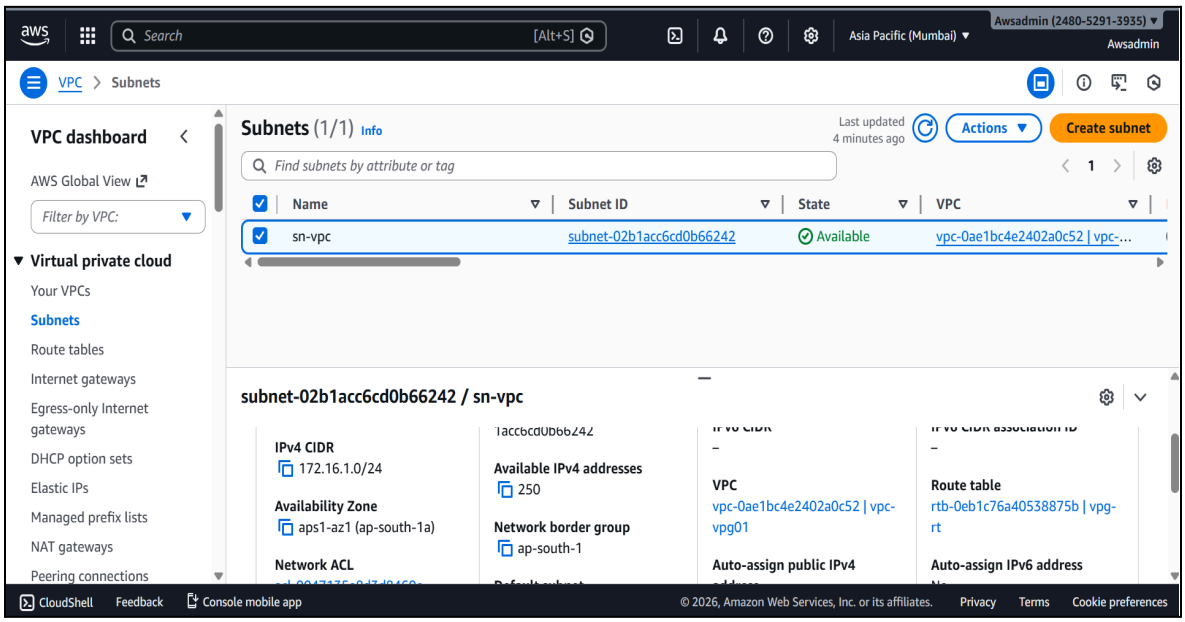
→ create a subnet.

Subnet > search with vpc.id

Name – vpc-sn01

Ipv4 – 172.16.1.0/24

Create ✔️



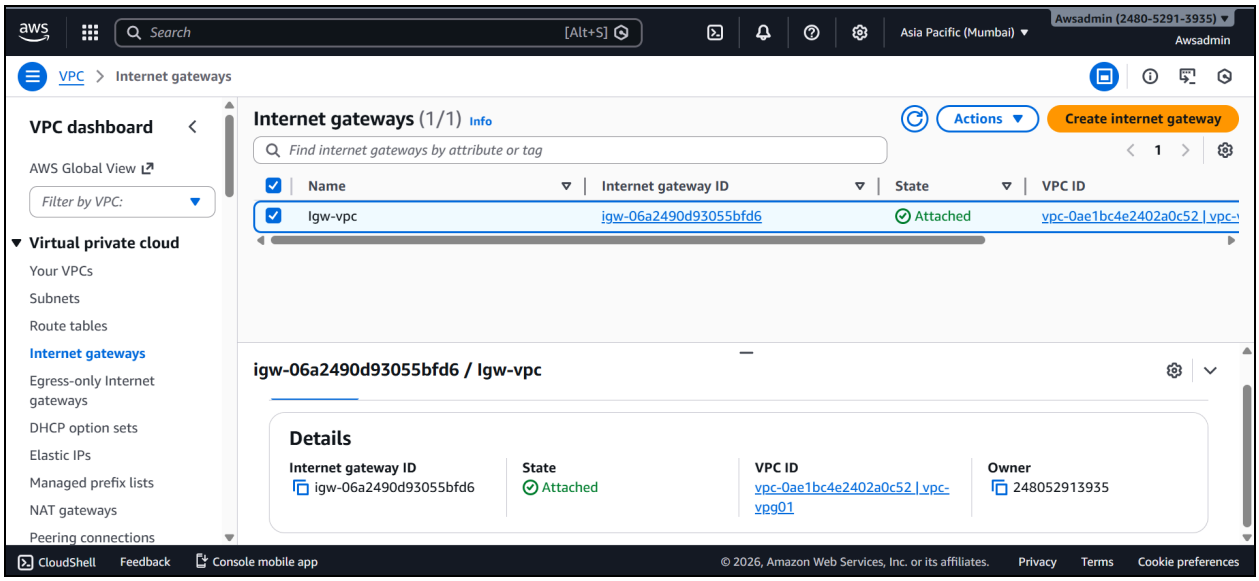

fig(6) created a subnet.

→ now configure the internet gateway.

Vpc > internet gateway > create

Name – vpg-igw

Create ✔️




fig(7) configured the internet gateway.

→ Now write the route table.

Vpc > route table > search with vpc id

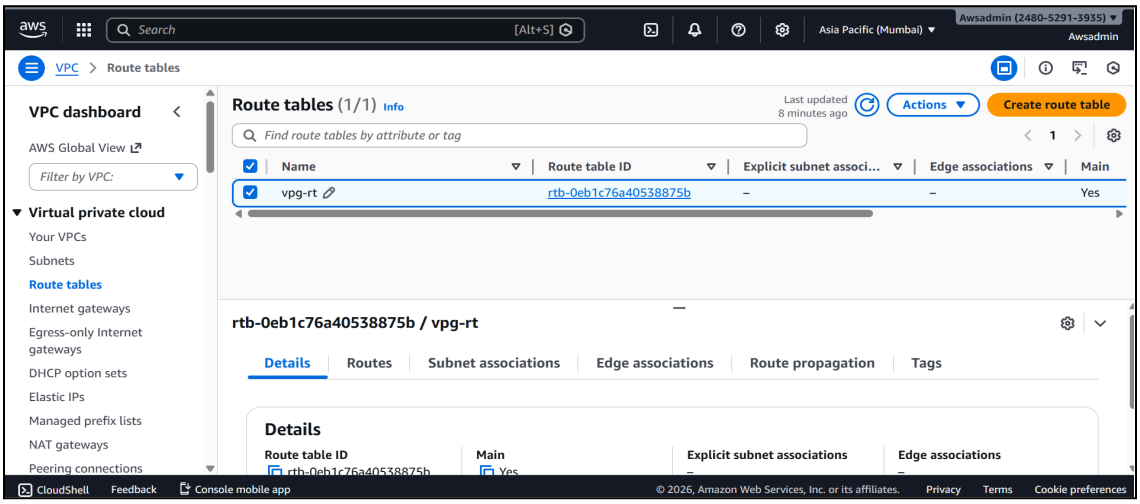
Edit name – vpg-rt.

>routes > edits routes > add route

any(0.0.0.0/0) - destination.

Target – igw

Save ✔️




fig(8) created a route table.

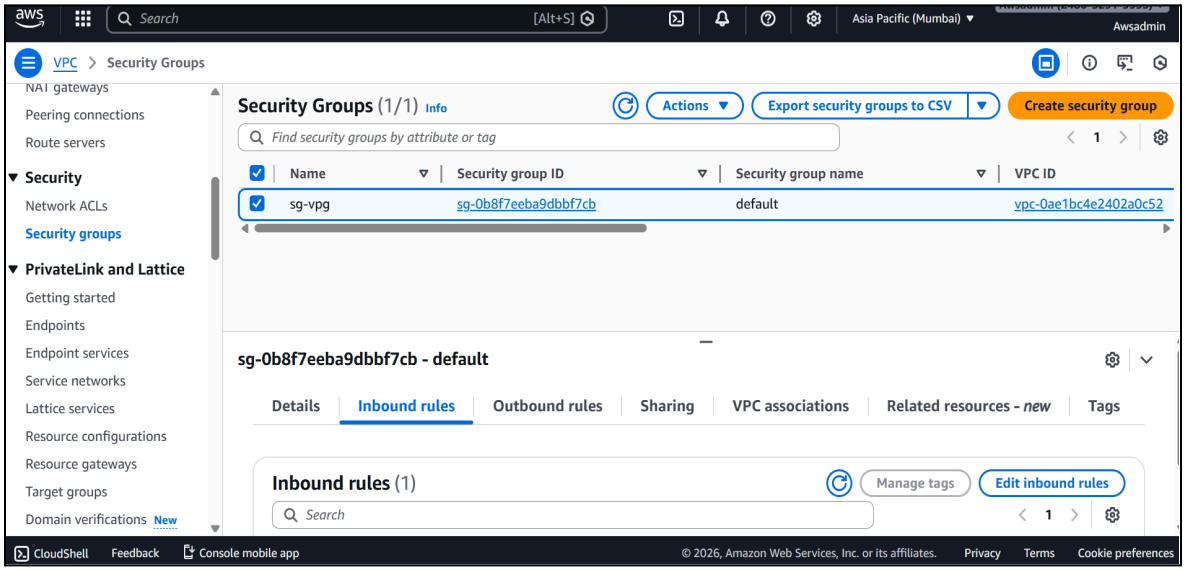
→ Now create the security groups

Search with vpc-id > edit name > sg-vpg

Inbound rules > edit inbound rules > delete.

Add rules > type-all traffic > source-anywhere ipv4.

Save ✔️




fig(9) configured the security groups.

→ now create a vm (ec2) in aws.

Search ec2 > ec2 > instance > launch instances

Name – awsvm01

Image – ubuntu image

Instance type – 2cpu 1gb

Key pair (login) :

Create new key pair

Name – vpg key

Type – RSA

File format – .pem

Create ✔️ (the key will be downloaded)

Networking setting :- edit

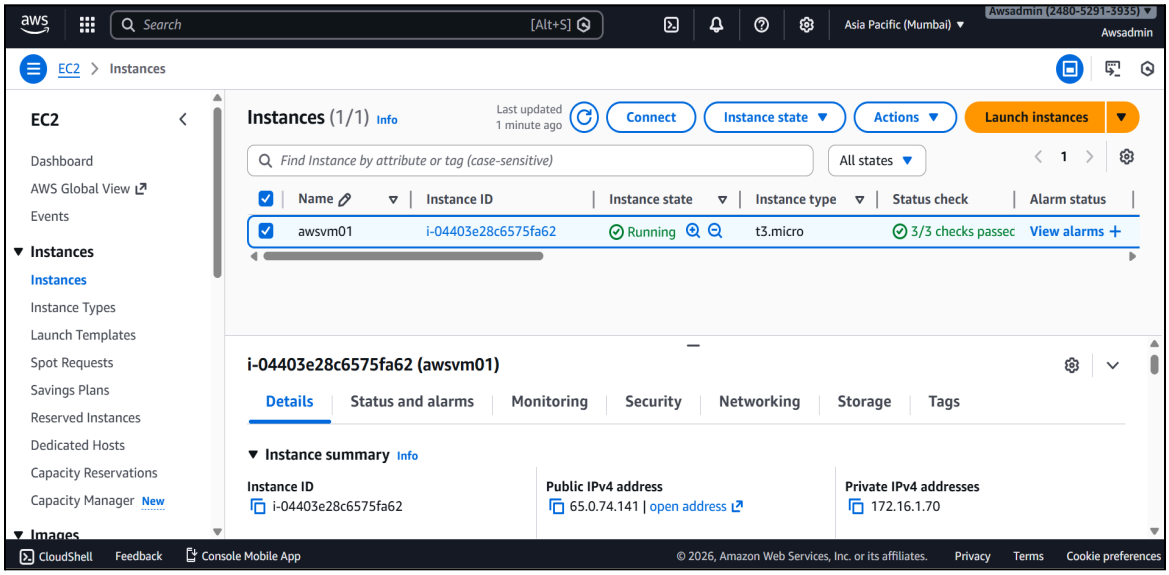
Enable the public ip, because we need to login to the machine.

Auto assign pub-ip – enable.

firewall(sg) – existing sg

Common sg – default sg.

Launch instance ✔️




fig(10) virtual machine is created.

→ Now create VPN :-

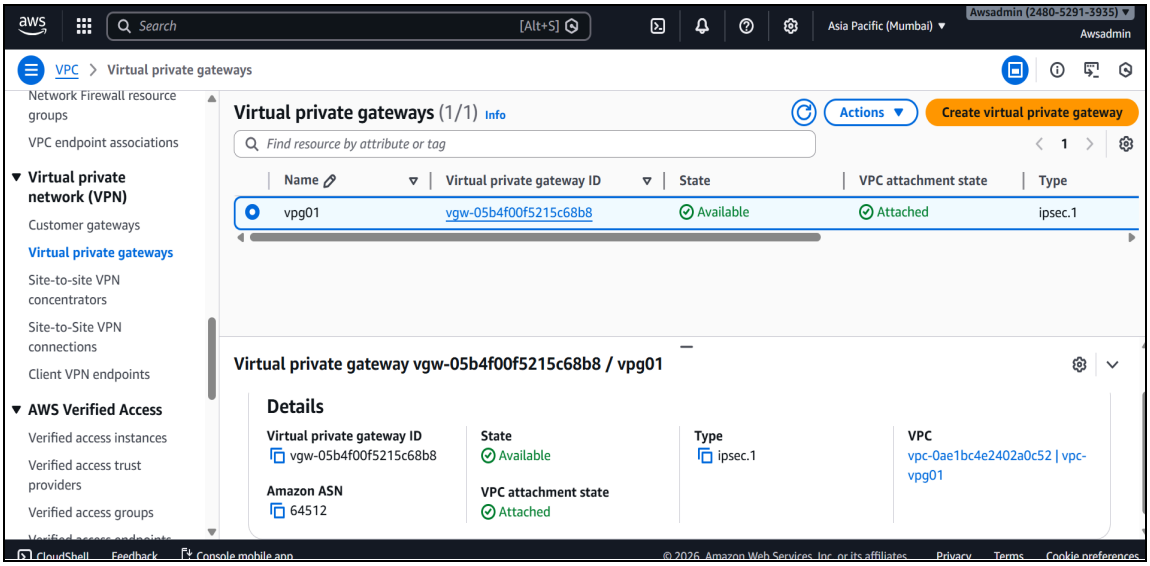
Search vpc > vpn > virtual private gateway.

Name – vpg01

Create✔️

After creating attach to vpc

Action > attach to vpc > available vpc.




fig(11)successfully created vpg.

Till here,

In azure - vm & vng is created.

In aws - vm & vpg created.

Now we need to configure the connection between two vpn's.

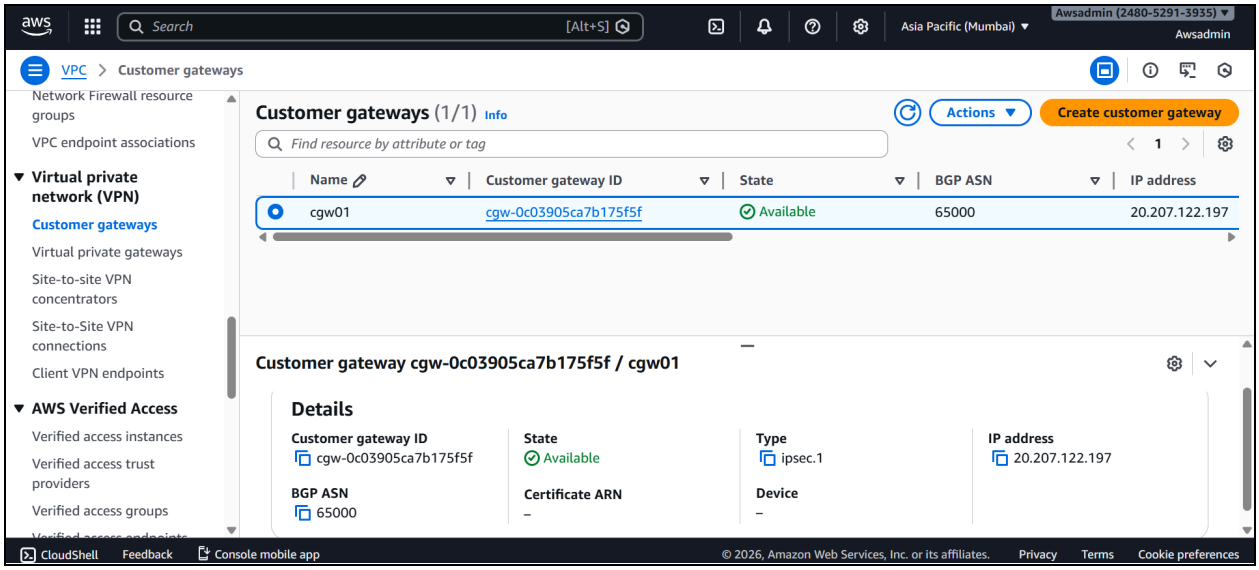
→ Now configure the customer gateways.

Vpc > vpn > customer gateways > create customer Gateway.

Name – cgw01

Ip address – we need to give azure vpn pub-ip address (vpn01).

Create ✔️




fig(12) configured customer gateways.

→ now configure the site to site connection.

Vpc > vpn > site to site vpn connection >

Create vpn connection.

Name – aws to azure

Target gateway type – vpg

Vpg – select available

Cgw – existing

Routing option – static(prefix ip)

(here, we need to use azure vn pvt-ip range)

Create ✔️

In aws for any traffic we need to write the route table. Without route it won't understand.

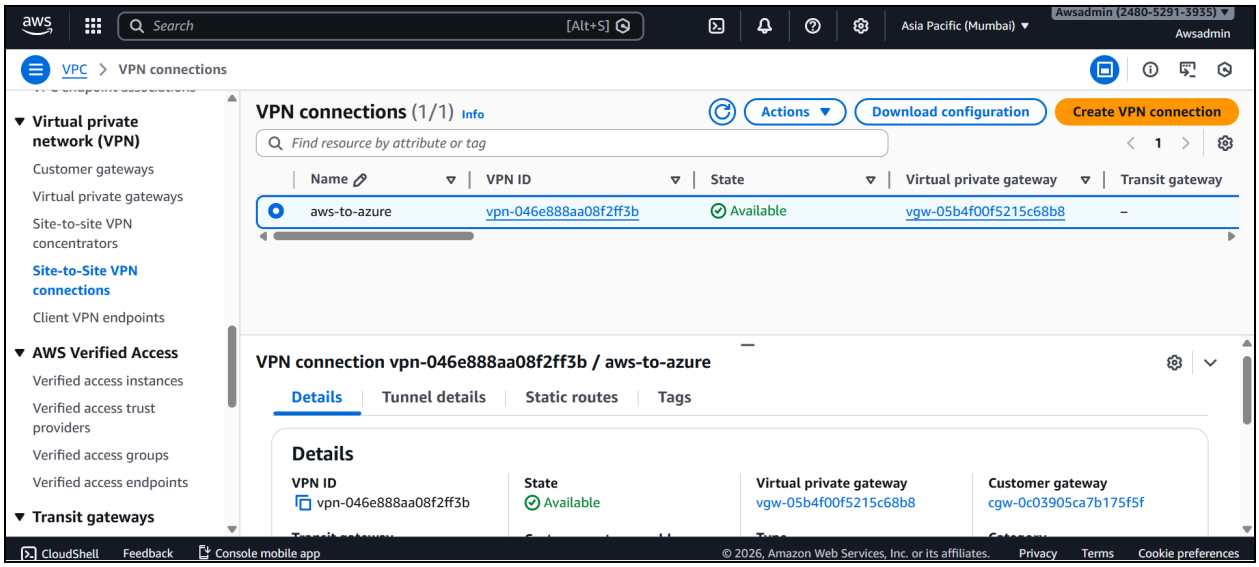
Vpc > your vpc > select vpg-vpc01.

Routes > edit routes >

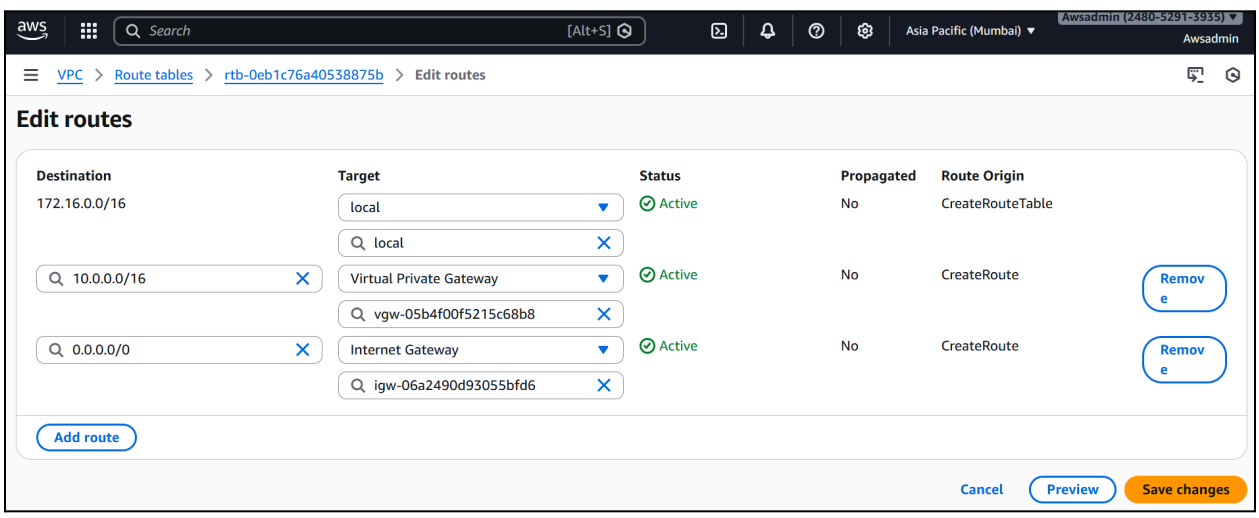
Destination – vnet pvt address

Target – vpg

(here if the machine is sending traffic with vnet pvt-ip then, it should use vpg.)




fig(13) configured site to site vpn connections.




fig(14) configured the route tables.

Goto site to site vpn connection

Download the configuration

Vendor – cisco system.inc(shows details in notepad)

Now goto <u>AZURE</u> :-

Search local n/w gateway > +create

Rg – vpn-rg

Region – c.i

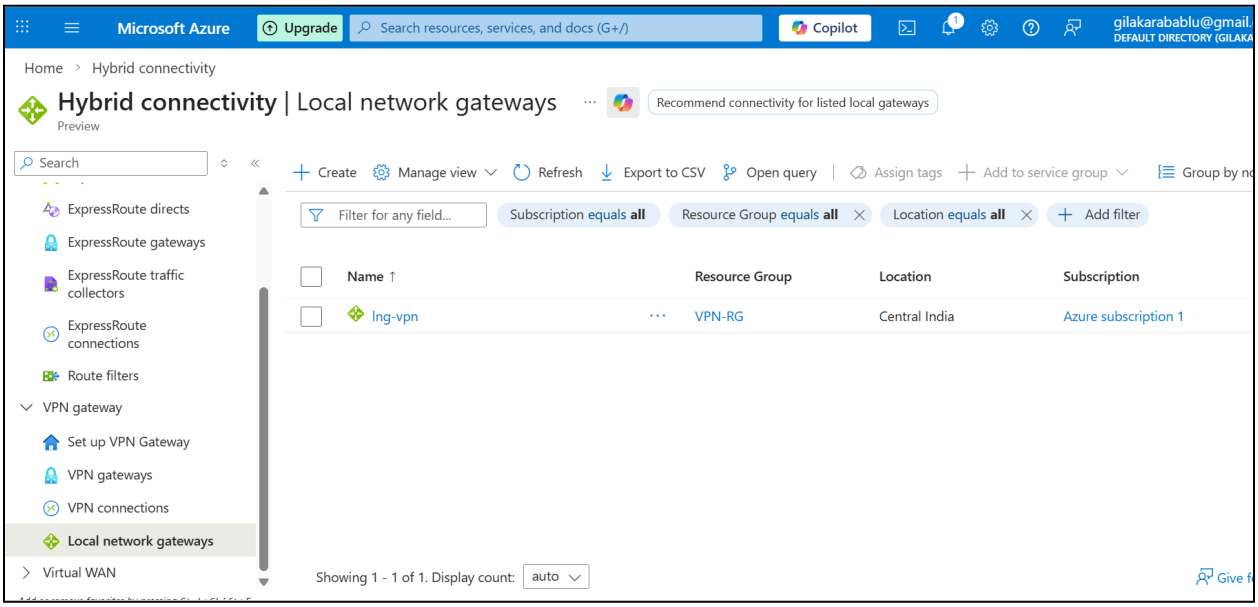
Name – lng-vpn

Ip address - give vpn public ip of aws

(copy from notepad)

Address space – aws private address range.

Create ✔️




fig(15) created a local network gateway.

Now configure the connection

Vng > vpn gateways > vpn01

Setting > connection > +add

Rg – vpn-rg

Connection type – site to site

Name – azure-to-aws.

Settings:

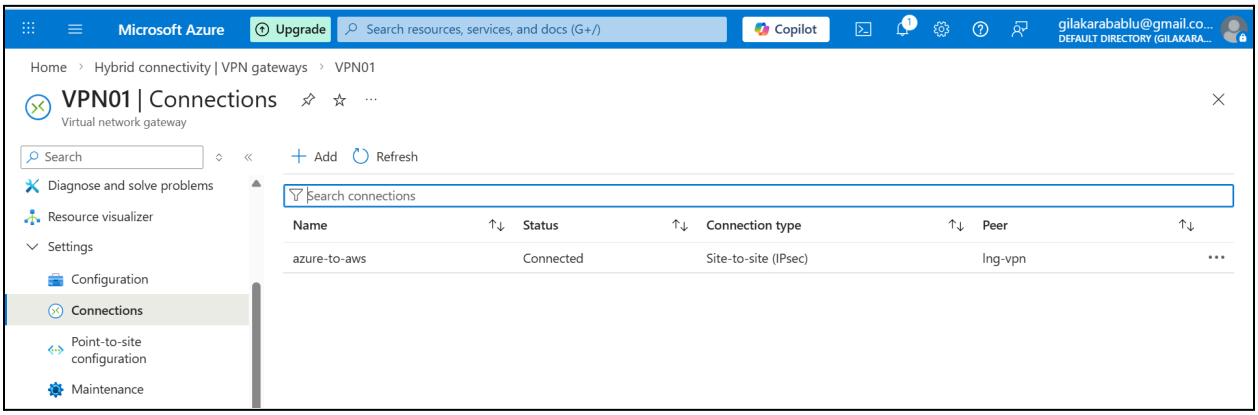
Vng – vpn01

Lng – lng-vpn

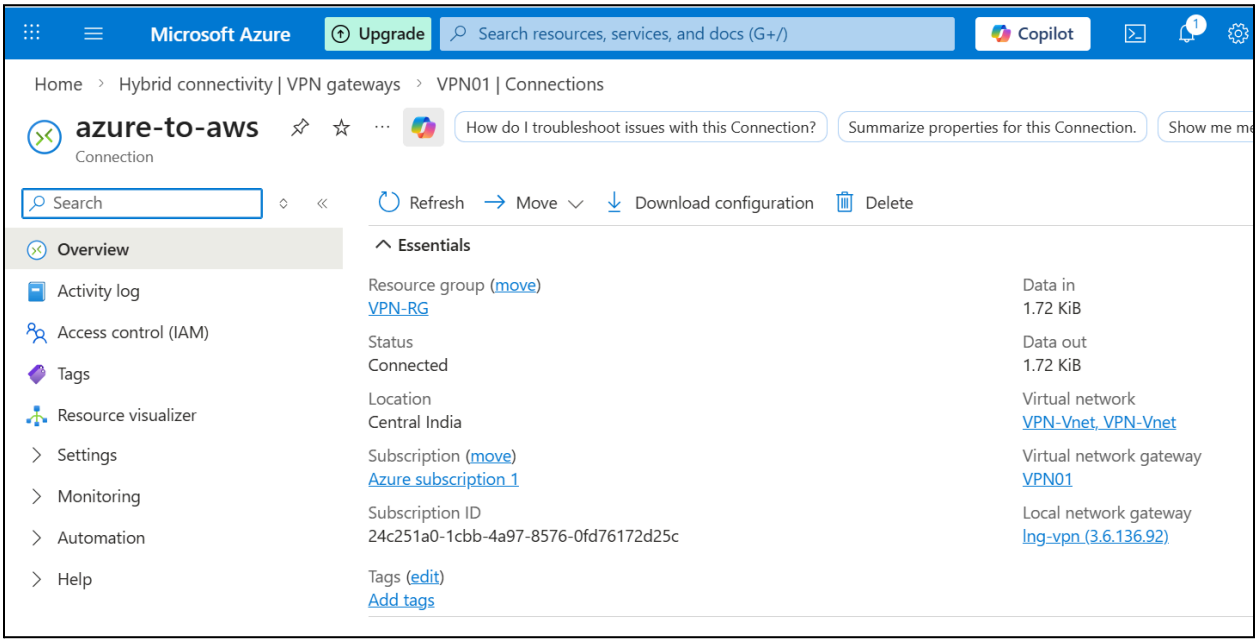
Authentication – shared key

(copy&paste pre shared key from notepad.)

Create. ✔️




fig(16) added vpn connection.




fig(17) configured the azure to aws connection.

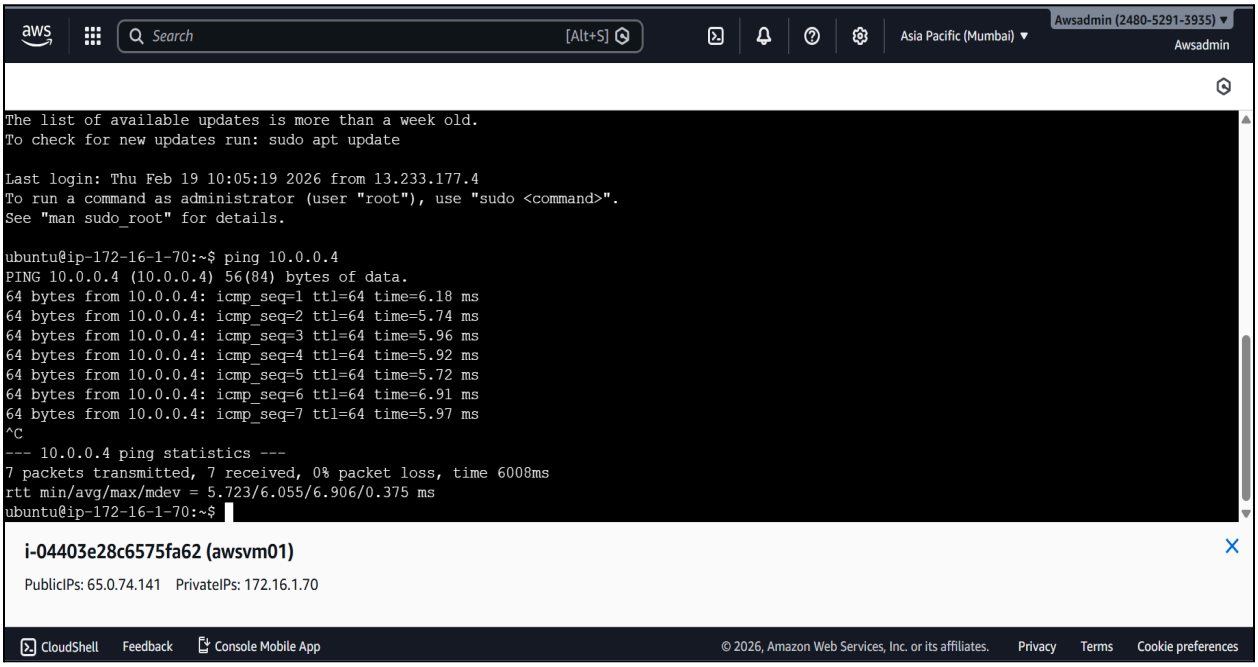
Now goto AWS

EC2 > open the console(near top right). i.e connect to the instance.

Execute the command:

#ping az vnet pvt-ip

We can see the communication is started between the machines, hence they are connected privately.




fig(18) successfully connected two machines using vpn.